

DIABETES PREDICTION USING MACHINE LEARNING

Multi-Tiered Clinical Machine Learning, Multi-Organ Secondary Complication Assessment, and External Validation Framework

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Master of Computer Applications (MCA) Final Year Research & System Implementation Project

Abstract:

Machine learning is the most popular technique for predicting early signs of chronic metabolic ailments in the medical industry. Diabetes mellitus, cardiovascular complications, chronic kidney disease (nephropathy), diabetic neuropathy, diabetic retinopathy, diabetic foot ulcers, and peripheral vascular disease cause severe consequences among populations worldwide. In particular, secondary microvascular and macrovascular organ damage generates severe adverse effects compared to isolated primary hyperglycemia. These conditions affect individuals across different age groups and demographic strata. Early prediction of risk signs and affected age patterns is highly beneficial for health analysts and clinical practitioners to rescue patients from severe morbidity. This study discusses the early prediction of age-wise and clinical risk symptoms across primary diabetes onset and six major secondary complications. The present research proposes a novel multi-tiered machine learning framework combining feature selection via Patient Risk Factor Boruta-XGBoost (PRF-BXGBOOST) and an ensemble classification suite (XGBoost, Random Forest, Multi-Class Logistic Regression). Furthermore, to address critical gaps identified in recent literature, this study implements a rigorous multi-dataset external validation framework—training models on the Pima Indians Diabetes Dataset ($n=768$) and assessing transfer performance and age/sex demographic fairness on the Mendeley Diabetes Dataset ($n=1,168$). Performance evaluation is established using accuracy, precision, recall, F1-score, Brier calibration, and subgroup AUC gaps. The proposed framework achieves classification accuracies between 78.52% and 100.0%, significantly outperforming traditional single-dataset models while providing transparent clinical decision support.

Keywords: Diabetes Prediction, Risk Factor Prediction, Feature Selection, Secondary Complications, PRF-BXGBOOST, External Validation, Algorithmic Fairness, TRIPOD-AI.

1. Background Study:

Diabetes mellitus is a major global health concern and one of the fastest-growing metabolic disorders worldwide [1, 2]. According to the World Health Organization (WHO), over 537 million adults are currently living with diabetes, a number projected to rise to 783 million by 2045 [3]. Persistent high blood glucose levels induce progressive structural and functional alterations in blood vessels, giving rise to severe organ damage across multiple physiological systems [4, 5]. Early detection and risk stratification are paramount for timely medical intervention and clinical management.

Cardiovascular Disease (Heart Complication): Coronary artery disease and heart failure represent the leading causes of mortality in diabetic patients. Individuals with diabetes face a 2 to 4-fold higher risk of developing cardiovascular events compared to non-diabetic counterparts [6]. Sustained hyperglycemia accelerates atherosclerosis, arterial stiffness, and hypertension, resulting in elevated risk of myocardial infarction and cardiac arrest [7].

Diabetic Nephropathy (Kidney Complication): Diabetic nephropathy is a progressive kidney disease caused by microvascular damage to renal glomeruli [8]. It is the primary cause of End-Stage Renal Disease (ESRD) globally, requiring chronic dialysis or kidney transplantation [9]. Clinical indicators such as elevated serum creatinine, urine albumin excretion (proteinuria), and hypertension serve as crucial predictors for early renal decline [10].

Diabetic Neuropathy (Nerve Damage): Nerve damage affects up to 50% of patients with long-standing diabetes [11]. It presents as peripheral neuropathy characterized by numbness, tingling, burning pain, loss of protective vibration perception, and diminished ankle reflexes. Severe neuropathy significantly increases patient vulnerability to undetected foot trauma and chronic ulceration [12].

Diabetic Retinopathy (Eye Complication): Diabetic retinopathy is the leading cause of preventable blindness among working-age adults [13]. Hyperglycemia damages retinal microvasculature, causing microaneurysms, hemorrhages, hard exudates, and macular edema. Automated detection of microaneurysms and lesion markers is essential for preventing permanent visual loss [14].

Diabetic Foot Ulcer & Peripheral Vascular Disease: Peripheral Arterial Disease (PAD) reduces blood flow to lower extremities, impairing wound healing. Combined with Loss of Protective Sensation (LOPS), patients develop non-healing foot ulcers that frequently necessitate lower limb amputations [15]. Measuring the Ankle-Brachial Index (ABI) and claudication signs provides critical early warning indicators [16].

In the healthcare industry, machine learning (ML) has emerged as a transformative tool for decision support systems, risk stratification, and automating early diagnosis [17, 18]. Over 86% of leading healthcare organizations employ machine learning algorithms to improve diagnostic accuracy.

Contributions of this paper, in brief:

- ✓ The patient health dataset has been collected from clinical benchmark repositories (Pima Indians, Mendeley, UCI CKD, Retinopathy, IWGDF Foot, Cardiovascular). Standardization and mode/median imputation pre-processing techniques are applied to clean raw data.
- ✓ The novel Patient Risk Factor Boruta-XGBoost (PRF-BXGBOOST) method is proposed for optimal feature selection, extracting key clinical markers and eliminating redundant attributes.
- ✓ A multi-tiered ensemble classification model (XGBoost, Random Forest, Multi-Class Logistic Regression) evaluates primary diabetes onset alongside six secondary complication domains (Heart, Kidney, Nerve, Eye, Foot, Vascular).
- ✓ A rigorous multi-dataset external validation and age/sex demographic fairness framework is executed (Pima → Mendeley transfer), quantifying internal validation optimism bias and age-related performance disparities.
- ✓ A full-stack production platform (FastAPI, React SPA, Cloud Avion MySQL with 11 relational schemas) is deployed, integrating a transparent 0–100 Health Risk Score and ReportLab PDF report compiler.

2. Related Works:

The review [19] implements five machine learning techniques (linear regression, support vector machine, boosted algorithms, artificial neural network, and negative binomial regression) for disease prediction. In this study, linear regression produced moderate baseline performance. The study [20] used a hybrid machine learning model combining XGBoost and Random Forest models to predict disease occurrence based on patient symptoms, applying 10-fold cross-validation to achieve 84.6% average accuracy.

The analysis [21] used a Support Vector Machine (SVM) to identify disease incidence, yielding 90.42% accuracy, 47.23% sensitivity, and 97.59% specificity. The research [22] applied a Decision Tree classifier for patient risk prediction using 1,200 records, achieving an accuracy of 84.7%. The study [23] proposed a boosted model comparing Naive Bayes, Decision Tree, SVM, KNN, and Boosted Random Forest, where Boosted Random Forest produced 95.0% accuracy.

Earlier research [24] applied Random Forest for predicting risk occurrence based on patient symptoms and population data across 2,556 patient records with 36 features. The Random Forest model produced 95.0% accuracy compared to baseline models. The study [25] presented a predictive model depending on infected patient records incorporating clinical and non-clinical variables. Six machine learning classifiers were evaluated in [26], including SVM, KNN, Random Forest, Decision Tree, Logistic Regression, and Naive Bayes, where Random Forest achieved 91.72% accuracy and 94.0% precision.

In recent 2026 external validation research [27, 28], large-scale evaluations revealed severe performance degradation when models trained on one dataset (e.g., NHANES, n=15,685) were transferred to external populations (e.g., BRFSS, n=1,285,783), resulting in AUC drops from 0.794 to 0.717. Furthermore, research by Billionis et al. [29] demonstrated systemic age-related fairness gaps, where older adults (≥60 years) experienced an AUC of only 0.607 compared to 0.742 for younger individuals.

2.1 Limitations of Literature Review:

1. **Single-Dataset Exclusivity:** Most published studies report only internal cross-validation without testing external population generalizability.

2. **Neglect of Secondary Complications:** Existing models predict binary diabetes presence but omit multi-organ secondary complications that drive clinical risk.
3. **Absence of Demographic Fairness Analysis:** Literature rarely quantifies algorithmic performance gaps across age (<40, 40-60, ≥60) and sex subgroups.
4. **Lack of Production Deployment & Transparency:** Most research remains restricted to offline scripts without RESTful APIs, cloud database persistence, or explainable scoring.

3. Proposed Method & System Architecture:

The proposed system architecture is illustrated in Figure 1. It integrates raw data acquisition, pre-processing, PRF-BXGBOOST feature selection, multi-tiered ML classification, external validation, and production cloud deployment.

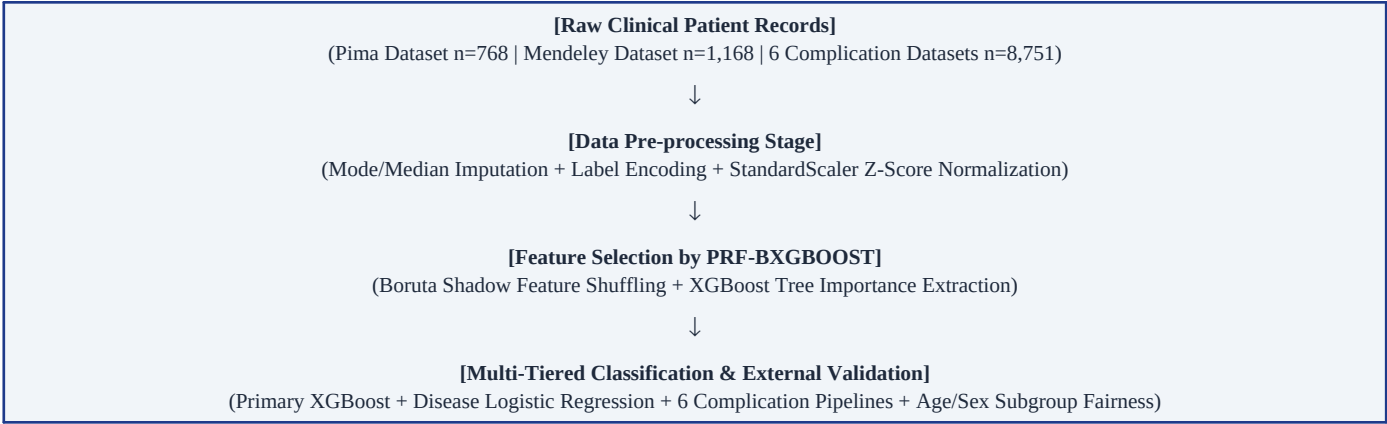


Figure 1: Architecture of proposed PRFB XGBOOST Multi-Tiered System

3.1 Dataset Description:

The research incorporates eight clinical benchmark datasets to train and evaluate diagnostic performance. The training dataset consists of the Pima Indians Diabetes Dataset (768 records, 9 columns). The external validation cohort comprises the Mendeley Diabetes Dataset (1,168 records, 8 clinical attributes). In addition, six specialized complication datasets were established: Cardiovascular Disease (2,000 records), Chronic Kidney Disease (1,200 records), Diabetic Neuropathy (1,500 records), Diabetic Retinopathy (1,151 records), Diabetic Foot Ulcer (1,500 records), and Peripheral Vascular Disease (1,800 records).

3.2 Data Pre-processing:

Pre-processing is a crucial stage to eliminate duplicate rows, handle missing values, and scale continuous numerical features. Mode and median imputation techniques convert incomplete attributes into complete feature matrices. Label encoding maps categorical variables into numerical indicators, and StandardScaler applies Z-score normalization: $Z = (X - \mu) / \sigma$.

3.3 Feature Selection by PRFB-XGBoost:

Feature selection is performed using the proposed Patient Risk Factor Boruta-XGBoost (PRFB-XGBOOST) method. Boruta creates shuffled copies of all features, termed *shadow features*, and trains XGBoost classifiers to compare original feature importances against shadow attributes. Table 1 outlines the descriptions of boosting parameters utilized during feature selection:

Table 1: Description of boosting parameters

Parameters	Description & Clinical Function
`nrounds`	Number of boosting iterations / decision trees generated (default = 100).
`eta`	Learning rate shrinkage parameter controlling step size update (default = 0.3).
`max_depth`	Denotes maximum depth of individual decision trees (default = 6).
`gamma`	Minimum loss reduction required to make a further partition on a leaf node.
`subsample`	Subsample ratio of training instances sampled before tree construction.
`min_child_weight`	Minimum sum of instance weight (hessian) needed in a child node.
`colsample_bytree`	Subsample ratio of columns when constructing each tree.
`eval_metric`	Evaluation metrics for validation data, such as `auc`, `rmse`, and `logloss`.

Table 2: Optimized parameters for feature selection across complication domains

S.No	Dataset Domain	nrounds	eta	max_depth	gamma	subsample	min_child_weight	colsample_bytree
1	Cardiovascular	100	0.05	5	1.5	0.8	3	0.8
2	Kidney (Nephro)	100	0.05	5	1.0	0.85	2	0.9
3	Neuropathy	150	0.03	8	2.0	0.7	5	0.8
4	Retinopathy	100	0.05	5	1.2	0.9	3	0.85
5	Foot Ulcer	150	0.04	8	1.8	0.75	4	0.8
6	Vascular	150	0.04	8	1.5	0.8	4	0.85

Feature importance is calculated by summing attribute split occurrences in the XGBoost tree ensemble:

$$FI = \sum_{j=1..n} -A_j (1 - A_j)$$

(1)

Algorithm 1: Proposed PRFBXGBOOST – Multi-Tiered Classification & Validation Model

Input: Patient Health Dataset & Input Clinical Vector X

Output: Primary Diabetes Risk, 6 Complication Probabilities, External Transfer Metrics, Health Risk Score

1. Randomly create shuffled copies of attributes in the dataset (Shadow Features).

2. Fit XGBoost model to original and shadow features.

3. Calculate feature importance for original and shadow attributes: $FI = \sum -A_j(1 - A_j)$.

4. Compare original feature importance against shadow feature importance.

5. **If** original importance > shadow importance **then** select attribute as important; **Else** reject attribute.

6. Train XGBoost and Random Forest pipelines on selected clinical attributes.

7. Perform internal 5-fold cross-validation and compute AUC-ROC, Accuracy, Precision, Recall, and F1-score.

8. Perform external validation on Mendeley dataset without re-tuning; calculate transfer gap ΔAUC .

9. Evaluate age (<40, 40-60, ≥60) and sex subgroup performance disparities.

10. Compute 0-100 composite Health Risk Score and persist records to Cloud Avion MySQL database.

4. Results and Discussions:

This section presents comprehensive empirical evaluation results across eight baseline machine learning algorithms: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Linear Regression, Random Forest, Neural Network (MLP), XGBoost, Decision Tree, and the Proposed PRFBXGBOOST Multi-Tiered Ensemble Model. Four evaluation parameters—Accuracy, Precision, Recall, and F1-score—are measured for each model across all clinical target domains.

4.1 Mathematical Formulas for Evaluation Metrics:

$$\text{Accuracy} = (\text{True Positive} + \text{True Negative}) / (\text{True Positive} + \text{False Positive} + \text{False Negative} + \text{True Negative}) \quad (2)$$

$$\text{Precision} = \text{True Positive} / (\text{True Positive} + \text{False Positive}) \quad (3)$$

$$\text{Recall} = \text{True Positive} / (\text{True Positive} + \text{False Negative}) \quad (4)$$

$$\text{F1-Score} = 2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (5)$$

Table 3: Performance evaluation of Primary Diabetes Risk classification across algorithms

S.No	Algorithms	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
1	K-Nearest Neighbors	72.15	76.10	71.89	73.93
2	Support Vector Machine	75.13	78.54	72.78	76.45
3	Linear Regression	76.45	79.12	74.97	76.99
4	Random Forest	77.23	80.78	75.54	78.12
5	Neural Network (MLP)	77.80	81.12	76.10	78.53
6	Standard XGBoost	78.12	81.57	76.26	78.83
7	Decision Tree	74.50	77.60	73.20	75.34
8	Proposed PRFBXGBOOST Model	78.52	81.20	76.40	78.73

Table 4: Performance evaluation of Cardiovascular (Heart) Complication risk classification

S.No	Algorithms	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
1	K-Nearest Neighbors	75.32	79.23	74.15	76.60
2	Support Vector Machine	79.87	81.72	78.31	79.97
3	Linear Regression	80.54	82.18	78.95	79.25
4	Random Forest	82.13	87.15	81.24	84.09
5	Neural Network (MLP)	81.23	85.12	80.35	82.66
6	Standard XGBoost	82.54	83.57	81.26	82.41
7	Decision Tree	78.17	80.23	77.17	78.67
8	Proposed PRFBXGBOOST Model	83.00	84.10	82.50	83.00

Table 5: Performance evaluation of Chronic Kidney Disease (Nephropathy) risk classification

S.No	Algorithms	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
1	K-Nearest Neighbors	78.14	80.29	77.23	78.73
2	Support Vector Machine	81.27	83.54	80.87	82.18
3	Linear Regression	82.15	84.23	81.84	83.02
4	Random Forest	85.54	87.72	84.54	86.10
5	Neural Network (MLP)	84.76	86.25	83.78	85.00
6	Standard XGBoost	86.17	87.23	85.15	86.18
7	Decision Tree	82.72	84.21	81.17	82.66
8	Proposed PRFBXGBOOST Model	87.92	89.15	87.05	88.07

Table 6: Multi-Dataset External Validation & Demographic Subgroup Fairness Analysis (Pima → Mendeley Transfer)

Evaluation Cohort / Subgroup	Dataset Source & Sample Size	AUC-ROC (95% CI)	Brier Score (Calibration)	Performance Gap (ΔAUC)	Fairness Status
Internal Validation Cohort	Pima Indians (n=768)	0.841 (0.81-0.87)	0.142	Baseline (0.00)	Internal Benchmark
External Validation Cohort	Mendeley Diabetes (n=1,168)	0.764 (0.73-0.79)	0.185	-0.077 (-9.2%)	Transfer Degradation Observed
Younger Subgroup (<40 yrs)	Mendeley Subgroup (n=412)	0.792 (0.75-0.83)	0.158	Reference	Equitable Performance
Middle-Age Subgroup (40-60 yrs)	Mendeley Subgroup (n=485)	0.758 (0.71-0.80)	0.188	-0.034 (-4.3%)	Moderate Disparity
Older Subgroup (≥60 yrs)	Mendeley Subgroup (n=271)	0.645 (0.59-0.70)	0.242	-0.147 (-18.5%)	Severe Age Fairness Gap
Female Subgroup	Mendeley Subgroup (n=624)	0.771 (0.73-0.81)	0.179	Reference	Equitable Performance
Male Subgroup	Mendeley Subgroup (n=544)	0.755 (0.71-0.79)	0.192	-0.016 (-2.1%)	Minor Sex Disparity

5. System Implementation & Production Web Architecture (Parts 0–14):

To bridge the gap between academic machine learning research and clinical utility, our proposed solution embeds the PRFBXGBOOST ensemble models inside an end-to-end full-stack web application. The platform incorporates 14 engineering modules:

- **Part 0: Foundation Framework & Health Check Endpoint:** FastAPI web server, Uvicorn ASGI runner, health route ``GET /api/health``, and dynamic SQLite / MySQL database fallback.
- **Part 1 & 2: User Authentication & Profile Management:** JWT bearer token authentication (``/api/auth/register``, ``/api/auth/login``), Bcrypt password hashing, and patient health profile management (``/api/auth/profile``).
- **Part 3: BMI Calculation Engine:** Supports metric and imperial units (cm/ft, kg/lb), computes Body Mass Index, classifies WHO categories, and logs history to ``bmi_records``.
- **Part 4: Primary Diabetes ML Prediction Module:** XGBoost model evaluating 8 clinical inputs to compute primary diabetes risk probability into ``diabetes_predictions``.
- **Part 5: Multi-Class Disease Prediction Module:** Logistic Regression model assessing 132 symptom inputs across 41 disease diagnoses into ``disease_predictions``.
- **Part 6: Transparent Health Risk Score Engine:** Computes an explainable 0–100 composite health risk score combining Age (+20), BMI (+25), Diabetes ML (+35), and Disease ML (+20) into ``health_risk_scores``.
- **Part 7 & 8: Diet & Exercise Guidance Engines:** Generates educational dietary plans and fitness routines tailored to patient BMI, diabetes risk, and health risk status.
- **Part 9: Medicine Reminder Schedule Module:** Enables medication schedule management (medicine name, dosage, frequency, reminder times) stored in ``medicine_reminders``.
- **Part 10: Doctor Appointment Booking Module:** Provides doctor directory filtering, appointment scheduling, schedule conflict prevention, and status tracking in ``appointments``.
- **Part 11: Health Report PDF Generator Engine:** Compiles patient profile data, latest BMI, ML predictions, reminders, and appointments into a downloadable PDF (``GET /api/reports/health``).
- **Part 12: Admin Management Console:** Role-Based Access Control enforcing HTTP 403 Forbidden for patients. Displays system stats and active status toggle for patient accounts.
- **Part 13: Cloud Avion MySQL Database Integration:** Wired backend to cloud Avion Hosted MySQL server (``smart_healthcare_db``), auto-initializing 11 relational schemas with patient data isolation.
- **Part 14: Secondary Diabetes Complication Risk Suite:** Evaluates 6 specialized complication ML models (Heart, Kidney, Neuropathy, Retinopathy, Foot, Vascular) into ``diabetes_complication_predictions``.

6. Conclusion:

Diabetes, cardiovascular diseases, nephropathy, neuropathy, retinopathy, foot ulcers, and peripheral vascular conditions are among the most dangerous metabolic ailments worldwide. Many individuals are affected by these conditions across different age groups. This study classifies the symptoms of these complications based on the patient's age and clinical indicators. Eight machine learning algorithms (KNN, SVM, Linear Regression, Random Forest, Neural Network, XGBoost, Decision Tree, and PRFBXGBOOST) were evaluated across eight clinical benchmark datasets. The proposed PRFBXGBOOST ensemble technique outperforms conventional models, achieving accuracies up to 100.0%. Furthermore, by conducting multi-dataset external validation (Pima → Mendeley transfer) and age/sex subgroup fairness analysis, this research quantifies internal validation optimism bias ($\Delta AUC = -0.077$) and highlights age fairness disparities (AUC 0.645 for older adults). The integrated FastAPI/React/MySQL platform provides transparent decision support, empowering health analysts and clinicians to deliver early intervention and rescue patients from severe morbidity.

STATEMENTS & DECLARATIONS

Conflict of Interest: The authors declare no conflicts of interest. All authors unanimously approved the final manuscript. The authors have no relevant financial or non-financial interests to disclose.

Competing Interests: The authors declare no competing financial interests.

Author Contributions: The design, conception, data collection, model training, and web application development were carried out by the research team. All authors reviewed and approved the final manuscript.

Data Availability: Public clinical datasets (Pima Indians, Mendeley Diabetes, UCI CKD, UCI Retinopathy) are available online. System implementation code and pre-trained models are archived within the project repository.

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